WEB-BASED ML MODEL EVALUATOR

TEST CASES DOCUMENT - companion to the Software Test Plan (STP) v1.0

1. Introduction

This document lists the individual test cases. The strategy, scope, schedule, roles and metrics are in the Software Test Plan (STP), which follows the course template. The cases cover all thirteen use cases (UC-01 to UC-13), the security requirements and the non-functional requirements, so that every requirement of the SRS v1.0 (MLE-F-001 to MLE-F-043, MLE-NF-001 to 005 and MLE-SR-001 to 009) has at least one test case, except MLE-SR-009 (TLS), which is a deployment check.

Requirement IDs: the earlier requirements document used FR-01 to FR-08 and NFR-01 to NFR-05. The SRS v1.0 replaces them with the MLE-F, MLE-NF and MLE-SR IDs; the SRS keeps the earlier ID next to each new one. Test case IDs have not changed.

2. Requirement Traceability

Every test case lists the SRS requirement it verifies. The table below is the reverse view: for each requirement, the test cases that cover it.

Requirement	Meaning	Covered by test cases	Count
MLE-F-001	Analyst registration	01_UT_01, 01_UT_02, 01_UT_03, 01_UT_04, 01_ST_01	5
MLE-F-002	Login, session and logout	02_UT_01, 02_UT_02, 02_IT_01, 02_IT_02, 02_ST_01	5
MLE-F-003	Admin accounts only through an Admin	SEC_02, 13_ST_01	2
MLE-F-010	Dataset upload and validation	03_UT_01, 03_UT_02, 03_UT_03, 03_UT_04, 03_IT_01, 03_IT_02	6
MLE-F-011	Model upload and validation	04_UT_01, 04_UT_02, 04_UT_03, 04_IT_01	4
MLE-F-012	Model and dataset compatibility check	04_IT_02, 04_IT_03, 06_IT_02	3
MLE-F-020	Task type selection and check	05_UT_01, 05_IT_01, 05_ST_01	3
MLE-F-021	Classification metrics	06_UT_01, 06_UT_03, 06_ST_01	3
MLE-F-022	Regression metrics	06_UT_02, 06_ST_02, 07_UT_01	3
MLE-F-023	Failed evaluation leaves nothing behind	06_IT_02, 06_ST_03	2
MLE-F-030	View metrics	07_UT_01, 07_UT_02, 07_IT_01	3
MLE-F-031	Visualizations	08_UT_01, 08_UT_02, 08_IT_01, 08_ST_01, 08_ST_02	5
MLE-F-032	Saved evaluations and history	06_IT_01, 07_IT_01, 10_UT_01, 10_UT_02	4
MLE-F-033	Compare evaluations	09_UT_01, 09_IT_01, 09_IT_02, 09_ST_01	4
MLE-F-034	PDF report	11_UT_01, 11_UT_02, 11_UT_03, 11_IT_01, 11_ST_01	5
MLE-F-040	Account list, suspend and reactivate	02_IT_03, 02_IT_04, 12_UT_01, 12_UT_02, 12_ST_01	5
MLE-F-041	Role change	13_UT_01, 13_ST_01	2
MLE-F-042	Audit log	13_IT_03	1
MLE-F-043	Protected accounts	12_IT_01, 12_IT_02, 13_IT_01, 13_IT_02	4

Requirement	Meaning	Covered by test cases	Count
MLE-NF-001	Performance (performance)	NF_PF_01, NF_PF_02, NF_PF_03	3
MLE-NF-002	Data isolation (security)	SEC_08, SEC_09, SEC_10, SEC_11, SEC_12	5
MLE-NF-003	Clear error reporting (reliability)	NF_ER_01, NF_ER_02, NF_ER_03, NF_ER_04	4
MLE-NF-004	First-time usability (usability)	NF_US_01, NF_US_02	2
MLE-NF-005	Extensibility (maintainability)	NF_EX_01	1
MLE-SR-001	Password hashing	SEC_01	1
MLE-SR-002	Login required	SEC_03	1
MLE-SR-003	Role enforcement	SEC_06, SEC_07	2
MLE-SR-004	CSRF protection	SEC_05	1
MLE-SR-005	Per-account isolation	SEC_08, SEC_09, SEC_10, SEC_11, SEC_12	5
MLE-SR-006	Safe login redirect	SEC_04	1
MLE-SR-007	Session security	02_IT_04	1
MLE-SR-008	Upload limits and storage	03_UT_01, 03_UT_04, 04_UT_01, 04_UT_03, NF_ER_04	5
MLE-SR-009	TLS 1.2+ when deployed (deployment check)	none (deployment check)	0

Note: compatibility must be checked before an evaluation (MLE-F-012, MLE-F-020); this is covered by 04_IT_02, 04_IT_03, 05_IT_01, 05_ST_01 and 06_IT_02.

3. Test Approach

Levels. The Test Case ID shows the level: **UT** = unit (one function or service on its own), **IT** = integration (several modules or the API together), **ST** = system (a full user workflow). The prefix is the use case number (01 = UC-01 Register, and so on). Cases that are not tied to one use case use **SEC** (security), **NF_PF** (performance), **NF_ER** (error handling), **NF_US** (usability) and **NF_EX** (extensibility).

Automation. Most cases are automated with pytest in `backend/tests`; the column 'Automated test' gives the exact test function. The suite contains additional supporting tests (for example hashing, e-mail case handling, duplicate-row counting) that are not listed here; they run together with the listed ones. Every test runs against its own temporary database and folders, so results do not depend on each other. 2 cases are manual (NF_PF_03 and NF_US_01) because they need a very large model file and a person who has not seen the app.

How to run. `cd backend; pip install -r requirements-dev.txt; python -m pytest tests -v`. For coverage (BL-15): `python -m pytest tests --cov=. --cov-branch --cov-report=term-missing --cov-report=html`.

4. Test Environment and Pass / Fail Criteria

Environment. Python 3.13 (the team may also use a newer version), Flask, pandas, scikit-learn, joblib, matplotlib, reportlab, SQLite. Sample files are in `backend/sample_data`.

Pass. The actual result equals the expected result. For MLE-NF-001 the evaluation finishes in under 10 seconds for up to 50,000 rows with one user (the SRS asks for 95 % of runs).

Fail. Any difference from the expected result. Each failure is logged as a defect with the test case ID and fixed under BL-13; the case is then run again.

Status of this document. The plan was executed on 4 November 2026 (BL-13). The results are in the columns 'Actual Result' and 'Test Result' and are summarised in section 7.

5. Test Cases

5.1 Security Validation

Authentication, role-based access, CSRF protection and isolation of each account's data (MLE-F-001 to 003, MLE-F-040 to 043, MLE-NF-002, MLE-SR-001 to 008).

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
SEC_01	MLE-SR-001	Passwords are stored hashed, never as plain text	App running with empty database.	Register an account; read the stored user record.	Password Str0ngPass!	Stored value is a hash and differs from the typed password.	`test_auth::test_passwords_are_stored_hashed`	As expected. 1 automated test(s) passed.	**Pass**
SEC_02	MLE-F-003	Registration can never create an Admin	Registration page open.	Submit the registration form with role=admin added to the request.	Valid details plus role=admin	Account is created as Analyst only.	`test_auth::test_01_IT_02_registration_cannot_create_an_admin`	As expected. 1 automated test(s) passed.	**Pass**
SEC_03	MLE-SR-002	Anonymous visitor cannot reach workspace pages or the API	Not logged in.	Open /dashboard, /models and /api/models without logging in.	None	Pages redirect to login; the API returns 401 JSON.	`test_auth::test_pages_redirect_anonymous_visitors_to_login` `test_auth::test_api_returns_401_json_for_anonymous_visitors`	As expected. 13 automated test(s) passed.	**Pass**
SEC_04	MLE-SR-006	Login cannot redirect to another website	Login page open.	Log in with next=https://other-site.example.	External next URL	User lands on an internal page, not the external site.	`test_auth::test_login_does_not_redirect_to_other_sites`	As expected. 3 automated test(s) passed.	**Pass**
SEC_05	MLE-SR-004	Request without a valid CSRF token is rejected	Logged in as Analyst.	Send a POST with no token, then with a wrong token; try the login form without a token.	Missing / wrong token	Each request is rejected and nothing changes.	`test_auth::test_post_without_csrf_token_is_rejected` `test_auth::test_post_with_wrong_csrf_token_is_rejected` `test_auth::test_login_form_without_csrf_token_is_rejected`	As expected. 3 automated test(s) passed.	**Pass**
SEC_06	MLE-SR-003	Analyst cannot open Admin pages or call the Admin API	Logged in as Analyst.	Open /admin/users and /admin/audit; call the admin API endpoints.	Analyst session	Access is refused; no data is returned.	`test_admin::test_analyst_cannot_use_admin_api` `test_admin::test_analyst_cannot_open_admin_pages`	As expected. 4 automated test(s) passed.	**Pass**
SEC_07	MLE-SR-003	Analyst cannot change roles or make themselves Admin	Logged in as Analyst.	Call the role and status endpoints for own and other accounts; try creating an account.	Analyst session	All attempts are refused; roles are unchanged.	`test_admin::test_analyst_cannot_change_roles_or_statuses_by_calling_the_api` `test_admin::test_analyst_cannot_make_themselves_admin` `test_admin::test_analyst_cannot_create_accounts`	As expected. 3 automated test(s) passed.	**Pass**
SEC_08	MLE-NF-002, MLE-SR-005	Analyst B cannot see Analyst A's files in any list	A has uploaded a model, dataset and evaluation; B is logged in.	As B, open the model, dataset and evaluation lists.	Two analyst accounts	B's lists show only B's own records.	`test_isolation::test_lists_only_show_your_own_records`	As expected. 1 automated test(s) passed.	**Pass**
SEC_09	MLE-NF-002, MLE-SR-005	Guessing another user's ID returns 404 (validate, delete, evaluate)	A owns model, dataset, evaluation; B is logged in.	As B, call validate, delete and evaluate using A's IDs.	A's record IDs	404 Not Found; A's data is untouched.	`test_isolation::test_bob_cannot_validate_alices_model_or_dataset_by_id` `test_isolation::test_bob_cannot_delete_alices_files` `test_isolation::test_bob_cannot_evaluate_with_alices_files`	As expected. 3 automated test(s) passed.	**Pass**
SEC_10	MLE-NF-002, MLE-SR-005	Guessing another user's evaluation, chart, comparison or report URL returns 404	A owns a completed evaluation; B is logged in.	As B, open A's evaluation page, chart, comparison and report URLs; include A's ID in B's own report request.	A's evaluation ID	404 Not Found; no part of A's data is included.	`test_isolation::test_bob_cannot_open_alices_evaluation_by_id` `test_isolation::test_bob_cannot_fetch_alices_charts_comparison_or_report` `test_isolation::test_a_report_cannot_smuggle_in_another_users_evaluation`	As expected. 3 automated test(s) passed.	**Pass**
SEC_11	MLE-NF-002, MLE-SR-005	Same file name used by two users does not clash or leak	A and B each upload a file with the same name.	List files as A and as B.	Same file name	Each user sees only their own file.	`test_isolation::test_uploaded_file_names_are_scoped_to_the_owner_in_listings`	As expected. 1 automated test(s) passed.	**Pass**

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
SEC_12	MLE-NF-002, MLE-SR-005	Admin cannot read an Analyst's data; compatibility results never mention other users' files	Analyst has uploaded data; Admin is logged in.	As Admin, request the Analyst's model, dataset and evaluation by ID; validate a file as another user.	Analyst record IDs	Not found; no other user's file names appear anywhere.	`test_isolation::test_admin_cannot_read_an_analysts_data` `test_isolation::test_cross_user_dataset_compatibility_report_never_mentions_other_users_files`	As expected. 2 automated test(s) passed.	**Pass**

5.2 Functional Test Cases by Use Case

UC-01 Register

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
01_UT_01	MLE-F-001	Blank required fields are rejected	Registration page open.	Leave required fields blank; submit.	Empty name / email / password	Each missing field is named; no account is created.	`test_auth::test_01_UT_01_blank_required_fields_are_rejected`	As expected. 1 automated test(s) passed.	**Pass**
01_UT_02	MLE-F-001	Invalid email and short password are rejected	Registration page open.	Submit once with a bad email, once with a password under 8 characters.	abc / name@; password 'short'	Clear message for each; no account is created.	`test_auth::test_01_UT_02_invalid_email_is_rejected` `test_auth::test_01_UT_02_short_password_is_rejected`	As expected. 5 automated test(s) passed.	**Pass**
01_UT_03	MLE-F-001	Duplicate email is rejected (ignoring letter case)	An account exists for a@x.com.	Register again with A@X.com.	A@X.com	'Account already exists' message; no second account.	`test_auth::test_01_UT_03_duplicate_email_is_rejected_ignoring_case`	As expected. 1 automated test(s) passed.	**Pass**
01_UT_04	MLE-F-001	Mismatched password confirmation is rejected	Registration page open.	Enter two different passwords; submit.	Passwords that differ	Rejected with a clear message.	`test_auth::test_mismatched_password_confirmation_is_rejected`	As expected. 1 automated test(s) passed.	**Pass**
01_ST_01	MLE-F-001	Valid registration creates an Analyst and signs in	Registration page open.	Fill valid details; submit.	Valid name, email, 8+ char password	Analyst account created and user lands on the dashboard.	`test_auth::test_01_IT_03_valid_registration_creates_an_analyst_and_signs_in`	As expected. 1 automated test(s) passed.	**Pass**

UC-02 Login

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
02_UT_01	MLE-F-002	Blank login fields are rejected	Login page open.	Submit with empty email or password.	Empty fields	Message asks for email and password.	`test_auth::test_02_UT_01_blank_login_fields_are_rejected`	As expected. 1 automated test(s) passed.	**Pass**
02_UT_02	MLE-F-002	Wrong password and unknown email give the same message	One Analyst account exists.	Log in with wrong password; log in with unknown email.	Wrong password / unknown email	Same generic message (does not reveal which part was wrong).	`test_auth::test_02_UT_03_wrong_password_and_unknown_email_give_the_same_message`	As expected. 1 automated test(s) passed.	**Pass**
02_IT_01	MLE-F-002	Analyst login opens the Analyst dashboard	Analyst account exists.	Log in as Analyst.	Valid Analyst credentials	Analyst dashboard opens.	`test_auth::test_02_IT_01_analyst_login_goes_to_the_analyst_dashboard`	As expected. 1 automated test(s) passed.	**Pass**
02_IT_02	MLE-F-002	Admin login opens the dashboard with Admin access	Admin account exists.	Log in as Admin.	Valid Admin credentials	Admin is signed in and can open Admin pages.	`test_auth::test_02_IT_02_admin_login_goes_to_the_admin_dashboard`	As expected. 1 automated test(s) passed.	**Pass**
02_IT_03	MLE-F-040	Suspended account cannot log in; works again once reactivated	Account suspended by Admin.	Try to log in; reactivate; log in again.	Suspended account	Login refused while suspended; succeeds after reactivation.	`test_auth::test_02_IT_03_suspended_account_cannot_log_in` `test_auth::test_reactivated_account_can_log_in_again`	As expected. 2 automated test(s) passed.	**Pass**
02_IT_04	MLE-F-040, MLE-SR-007	Suspending an account ends its open session at once	User is logged in; Admin suspends the account.	User clicks any page after suspension.	Open session	User is sent out immediately (no wait for session expiry).	`test_auth::test_suspending_an_account_ends_its_open_session_immediately`	As expected. 1 automated test(s) passed.	**Pass**
02_ST_01	MLE-F-002	Logout ends the session; login returns to requested page	Logged in.	Log out, then open a workspace page; log in again.	None	Logout blocks access; after login the user returns to the page asked for.	`test_auth::test_logout_ends_the_session` `test_auth::test_login_redirects_back_to_the_requested_page`	As expected. 2 automated test(s) passed.	**Pass**

UC-03 Upload Dataset

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
03_UT_01	MLE-F-010, MLE-SR-008	CSV accepted, other file types rejected	Logged in as Analyst.	Upload a .csv; then upload a .txt.	sample_classification.csv; notes.txt	CSV accepted; other type rejected naming the accepted type.	`test_dataset_service::test_03_UT_01_csv_dataset_is_accepted` `test_dataset_service::test_03_UT_01_unsupported_dataset_formats_are_rejected`	As expected. 5 automated test(s) passed.	**Pass**
03_UT_02	MLE-F-010	Corrupt dataset is rejected with a reason	Logged in.	Upload a .csv that is not valid CSV.	Corrupt file	Rejected with a specific reason; nothing stored.	`test_dataset_service::test_03_UT_02_corrupt_dataset_is_rejected` `test_api_workflow::test_03_IT_02_corrupt_dataset_is_rejected_with_a_reason`	As expected. 4 automated test(s) passed.	**Pass**
03_UT_03	MLE-F-010	Target column must be given and must exist	Logged in.	Upload without a target; upload with a target name not in the file.	Blank target; wrong target	Rejected; message lists the real column names.	`test_dataset_service::test_03_UT_03_target_column_must_be_specified` `test_dataset_service::test_03_UT_03_target_column_must_exist_and_error_lists_columns` `test_api_workflow::test_03_ST_02_dataset_upload_without_a_target_is_rejected`	As expected. 3 automated test(s) passed.	**Pass**
03_UT_04	MLE-F-010, MLE-SR-008	Row limit: 50,000 accepted, 50,001 rejected	Logged in.	Upload a file with exactly 50,000 rows; then one with 50,001.	50,000 and 50,001 rows	First accepted; second rejected with the limit named.	`test_dataset_service::test_dataset_exactly_at_row_limit_is_accepted` `test_dataset_service::test_dataset_over_row_limit_is_rejected`	As expected. 2 automated test(s) passed.	**Pass**
03_IT_01	MLE-F-010	Dataset with missing target values is marked invalid	Dataset uploaded.	Validate a dataset where some target cells are empty.	bad_dataset_missing_target.csv	Marked invalid with the reason; cannot be used for evaluation.	`test_dataset_service::test_missing_target_values_make_dataset_invalid`	As expected. 1 automated test(s) passed.	**Pass**
03_IT_02	MLE-F-010	Valid dataset passes validation with a schema summary	Dataset uploaded.	Validate a clean dataset.	sample_classification.csv	Marked valid; row count and columns shown; missing feature values are reported but allowed.	`test_dataset_service::test_clean_dataset_passes_validation_with_schema_summary` `test_dataset_service::test_missing_values_in_features_are_reported_but_still_valid`	As expected. 2 automated test(s) passed.	**Pass**

UC-04 Upload Model

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
04_UT_01	MLE-F-011, MLE-SR-008	Only .pkl and .joblib models are accepted	Logged in.	Upload a .pkl; upload a .txt.	sample_classifier.pkl; notes.txt	Supported format accepted; others rejected.	`test_model_service::test_04_UT_01_supported_model_formats_are_accepted` `test_model_service::test_04_UT_01_unsupported_model_formats_are_rejected`	As expected. 5 automated test(s) passed.	**Pass**
04_UT_02	MLE-F-011	No file or empty file is rejected	Logged in.	Submit with no file; submit an empty file.	No file / 0-byte file	Rejected with a clear message.	`test_model_service::test_04_UT_02_no_file_selected_is_rejected` `test_model_service::test_empty_model_file_is_rejected`	As expected. 2 automated test(s) passed.	**Pass**

04_UT_03	MLE-F-011, MLE-SR-008	Size limit: 100 MB accepted, above 100 MB rejected	Logged in.	Upload a file of exactly 100 MB; then one slightly larger.	100 MB and 100 MB + 1 byte	First accepted; second rejected with the limit named (HTTP 413 as JSON).	`test_model_service::test_model_exactly_at_size_limit_is_accepted`
							`test_model_service::test_model_over_size_limit_is_rejected`
							`test_api_workflow::test_oversized_upload_returns_413_json`
								As expected. 3 automated test(s) passed.	**Pass**

04_IT_01	MLE-F-011	Corrupt model is flagged invalid with a reason	Corrupt .pkl uploaded.	Validate the model.	Corrupt model file	Marked invalid with a specific reason.	`test_model_service::test_corrupt_model_file_is_marked_invalid`
							`test_model_service::test_loadable_object_without_predict_is_marked_invalid`
							`test_api_workflow::test_04_IT_03_invalid_model_file_is_rejected_immediately_with_a_reason`
								As expected. 3 automated test(s) passed.	**Pass**

04_IT_02	MLE-F-012	Model compatible with dataset passes; missing features fail	Valid dataset uploaded.	Check a matching model; check a model that needs a column the dataset lacks.	Matching / non-matching model	Compatible model accepted; the other rejected naming the missing columns.	`test_evaluation_service::test_04_IT_02_compatible_model_is_accepted`
							`test_evaluation_service::test_04_IT_02_model_with_features_missing_from_dataset_is_rejected`
							`test_evaluation_service::test_04_ST_02_model_with_wrong_feature_count_is_rejected`
								As expected. 3 automated test(s) passed.	**Pass**

04_IT_03	MLE-F-012	Compatibility with each existing file is shown at validation	Model(s) and dataset(s) uploaded.	Validate a model, then a dataset.	One model, two datasets	Result lists fit with each of the user's existing datasets (or models).	`test_compatibility_at_upload::test_validated_model_reports_fit_with_each_existing_dataset` `test_compatibility_at_upload::test_validated_dataset_reports_fit_with_each_existing_model`	As expected. 2 automated test(s) passed.	**Pass**

UC-05 Select Task Type

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result

05_UT_01	MLE-F-020	Classification and regression accepted; unknown task rejected	Valid model and dataset.	Select each task; submit an unknown task name.	classification / regression / 'cluster'	First two accepted; unknown rejected.	`test_evaluation_service::test_05_UT_02_classification_task_is_accepted`
							`test_evaluation_service::test_05_UT_03_regression_task_is_accepted`
							`test_evaluation_service::test_unknown_task_type_is_rejected`
								As expected. 6 automated test(s) passed.	**Pass**

05_IT_01	MLE-F-020	Task that does not match the model type is rejected	Classifier and regressor uploaded.	Choose regression for a classifier and classification for a regressor.	Mismatched pairs	Rejected before evaluation; no metrics shown.	`test_evaluation_service::test_05_IT_03_regression_task_with_a_classifier_is_rejected`
							`test_evaluation_service::test_05_IT_03_classification_task_with_a_regressor_is_rejected`
							`test_api_workflow::test_05_ST_02_mismatched_task_type_is_rejected_over_http`
								As expected. 3 automated test(s) passed.	**Pass**

05_ST_01	MLE-F-020	Task that does not match the dataset target is rejected	Dataset has a continuous or text target.	Choose classification for continuous target; regression for text target.	Mismatched target	Rejected with a reason.	`test_evaluation_service::test_05_ST_02_classification_task_with_a_continuous_target_is_rejected` `test_evaluation_service::test_regression_task_with_a_text_target_is_rejected`	As expected. 2 automated test(s) passed.	**Pass**

UC-06 Evaluate Model

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
06_UT_01	MLE-F-021	Classification metrics match hand calculation	Known model and dataset.	Evaluate; compare each metric with a hand-computed value.	Small fixed dataset	Accuracy, precision, recall, F1 (and ROC-AUC) equal the expected values.	`test_evaluation_service::test_06_UT_03_classification_metrics_match_hand_calculation` `test_evaluation_service::test_classification_result_contains_exactly_the_defined_metrics`	As expected. 2 automated test(s) passed.	**Pass**
06_UT_02	MLE-F-022	Regression metrics match hand calculation	Known model and dataset.	Evaluate; compare MAE, MSE, RMSE, R^2 with hand values.	Small fixed dataset	All four metrics equal the expected values.	`test_evaluation_service::test_06_UT_03_regression_metrics_match_hand_calculation` `test_evaluation_service::test_regression_result_contains_exactly_the_defined_metrics`	As expected. 2 automated test(s) passed.	**Pass**
06_UT_03	MLE-F-021	ROC-AUC skipped with a note when the model has no probabilities; multiclass supported	Model without predict_proba.	Evaluate a classification model that gives no probabilities; evaluate a multiclass one.	No-probability model; 3-class data	ROC-AUC omitted with an explanation; multiclass reports ROC-AUC.	`test_evaluation_service::test_roc_auc_is_skipped_with_a_note_when_model_has_no_probabilities` `test_evaluation_service::test_multiclass_classification_reports_roc_auc`	As expected. 2 automated test(s) passed.	**Pass**
06_IT_01	MLE-F-032	Completed evaluation is stored with its details	Valid model and dataset.	Run an evaluation; read the stored record.	Valid pair	Record holds model, dataset, task, metrics, time and owner.	`test_evaluation_service::test_06_IT_03_completed_evaluation_is_stored_with_its_metadata`	As expected. 1 automated test(s) passed.	**Pass**
06_IT_02	MLE-F-012, MLE-F-023	Unvalidated or invalid model/dataset cannot be evaluated; nothing is stored	File uploaded but not validated (or invalid).	Try to evaluate.	Unvalidated / invalid files	Rejected with a reason and no evaluation saved.	`test_evaluation_service::test_unvalidated_model_cannot_be_evaluated` `test_evaluation_service::test_unvalidated_dataset_cannot_be_evaluated` `test_evaluation_service::test_model_that_failed_validation_cannot_be_evaluated` `test_api_workflow::test_06_ST_02_evaluating_an_unvalidated_model_is_rejected_and_stores_nothing`	As expected. 4 automated test(s) passed.	**Pass**
06_ST_01	MLE-F-021	End-to-end classification workflow	Logged in as Analyst.	Upload dataset and model, validate, choose classification, evaluate, view results.	Sample classifier and dataset	Metrics and charts are shown.	`test_api_workflow::test_03_ST_01_04_ST_01_06_ST_01_classification_workflow_end_to_end`	As expected. 1 automated test(s) passed.	**Pass**
06_ST_02	MLE-F-022	End-to-end regression workflow; several models on one dataset	Logged in.	Run the regression workflow; evaluate several models on one dataset.	Sample regressor and dataset	Regression metrics shown; each model gets its own evaluation.	`test_api_workflow::test_regression_workflow_end_to_end` `test_api_workflow::test_several_models_can_be_evaluated_against_one_dataset`	As expected. 2 automated test(s) passed.	**Pass**
06_ST_03	MLE-F-023	Model that crashes while predicting gives a clear error; nothing stored	Model that raises an error on predict.	Evaluate.	Faulty model	Specific error shown; no evaluation saved; app keeps running.	`test_evaluation_service::test_06_ST_02_model_that_crashes_while_predicting_gives_a_clear_error_and_stores_nothing`	As expected. 1 automated test(s) passed.	**Pass**

UC-07 View Metrics

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
07_UT_01	MLE-F-022, MLE-F-030	Evaluation page shows the right metrics for the task	Completed evaluations exist.	Open a classification and a regression evaluation.	Saved evaluations	Metrics and charts shown; regression page shows only regression metrics.	`test_charts_compare_reports::test_07_UT_01_evaluation_page_shows_metrics_and_charts` `test_charts_compare_reports::test_regression_evaluation_page_shows_only_regression_metrics`	As expected. 2 automated test(s) passed.	**Pass**
07_UT_02	MLE-F-030	Missing evaluation gives a message, not a crash	Logged in.	Open an evaluation ID that does not exist.	Unknown ID	Clear 'not found' message.	`test_charts_compare_reports::test_07_UT_02_missing_evaluation_gives_a_message_not_a_crash`	As expected. 1 automated test(s) passed.	**Pass**
07_IT_01	MLE-F-030, MLE-F-032	Reopened evaluation has exactly the stored values	Completed evaluation saved.	Reopen it later.	Saved evaluation	Values equal those first shown.	`test_charts_compare_reports::test_07_IT_01_reopening_gives_exactly_the_stored_values`	As expected. 1 automated test(s) passed.	**Pass**

UC-08 View Visualizations

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
08_UT_01	MLE-F-031	Classification shows confusion matrix and ROC curve	Classification evaluation saved.	Open the evaluation charts.	Classifier with probabilities	Confusion matrix and ROC curve are drawn.	`test_charts_compare_reports::test_08_UT_01_classification_gets_confusion_matrix_and_roc_curve` `test_charts_compare_reports::test_08_UT_02_each_available_chart_renders_a_png`	As expected. 2 automated test(s) passed.	**Pass**
08_UT_02	MLE-F-031	Regression shows residual and predicted-vs-actual plots	Regression evaluation saved.	Open the evaluation charts.	Regressor	Both plots are drawn.	`test_charts_compare_reports::test_08_UT_03_regression_gets_residual_and_predicted_vs_actual_plots`	As expected. 1 automated test(s) passed.	**Pass**
08_IT_01	MLE-F-031	No chart of the wrong task type; unknown chart is 404	Evaluations of both types saved.	Ask for a regression chart on a classification evaluation and vice versa; ask for an unknown chart.	Wrong chart kinds	Not available / 404.	`test_charts_compare_reports::test_08_IT_03_no_classification_chart_for_regression_and_vice_versa` `test_charts_compare_reports::test_unknown_chart_kind_is_404`	As expected. 2 automated test(s) passed.	**Pass**
08_ST_01	MLE-F-031	No ROC curve when the model gives no probabilities	Classifier without predict_proba evaluated.	Open its charts.	No-probability model	Confusion matrix only; no ROC curve.	`test_charts_compare_reports::test_08_ST_02_no_roc_curve_when_the_model_gives_no_probabilities`	As expected. 1 automated test(s) passed.	**Pass**
08_ST_02	MLE-F-031	Chart data agrees with the results	Evaluations saved.	Compare confusion-matrix counts and ROC data with the predictions; check large regression sampling.	Fixed predictions; 50,000-row regression	Counts match; ROC is a valid curve; large charts are sampled but metrics use every row.	`test_charts_compare_reports::test_confusion_matrix_counts_match_the_predictions` `test_charts_compare_reports::test_roc_curve_data_is_a_valid_curve` `test_charts_compare_reports::test_multiclass_evaluation_gets_one_roc_curve_per_class` `test_charts_compare_reports::test_large_regression_charts_use_a_sample_but_metrics_use_every_row`	As expected. 4 automated test(s) passed.	**Pass**

UC-09 Compare Models

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
09_UT_01	MLE-F-033	Fewer than two evaluations cannot be compared	Evaluations exist.	Compare one evaluation; compare the same one twice; include a missing ID.	1 evaluation / duplicate / missing	Refused with a reason.	`test_charts_compare_reports::test_comparing_fewer_than_two_evaluations_is_refused` `test_charts_compare_reports::test_comparing_the_same_evaluation_twice_counts_as_one` `test_charts_compare_reports::test_comparing_with_a_missing_evaluation_is_refused`	As expected. 3 automated test(s) passed.	**Pass**
09_IT_01	MLE-F-033	Different task types are not compared and the user is warned	One classification and one regression evaluation.	Select both and compare.	Mixed task types	Warning shown; no comparison table.	`test_charts_compare_reports::test_09_IT_01_different_task_types_are_not_compared_and_the_user_is_warned`	As expected. 1 automated test(s) passed.	**Pass**
09_IT_02	MLE-F-033	Same-task evaluations are shown side by side	Two or more same-task evaluations.	Select them and compare.	Two classification evaluations	Metrics aligned side by side; comparison chart renders.	`test_charts_compare_reports::test_09_IT_02_same_task_evaluations_are_compared_side_by_side` `test_charts_compare_reports::test_compare_page_and_chart_render`	As expected. 2 automated test(s) passed.	**Pass**
09_ST_01	MLE-F-033	Comparison never picks a winner	Same-task evaluations compared.	Read the comparison output.	Any compared pair	Values shown with no ranking or 'best' label.	`test_charts_compare_reports::test_comparison_never_ranks_the_models`	As expected. 1 automated test(s) passed.	**Pass**

UC-10 View Evaluation History

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
10_UT_01	MLE-F-032	History is empty for a new user	New account.	Open the history page.	None	Empty-state message shown.	`test_charts_compare_reports::test_10_UT_03_history_starts_empty`	As expected. 1 automated test(s) passed.	**Pass**
10_UT_02	MLE-F-032	History lists newest first	Several evaluations saved.	Open the history page.	3 evaluations	Newest at the top; each links to its saved result.	`test_charts_compare_reports::test_10_UT_02_history_lists_newest_first`	As expected. 1 automated test(s) passed.	**Pass**

UC-11 Generate Report

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
11_UT_01	MLE-F-034	Single-evaluation PDF report	Evaluation saved.	Download the report for one evaluation.	One saved evaluation	Valid PDF with model, dataset, metrics and charts.	`test_charts_compare_reports::test_11_UT_01_single_evaluation_report_is_a_pdf` `test_charts_compare_reports::test_report_embeds_every_chart_as_an_image`	As expected. 2 automated test(s) passed.	**Pass**
11_UT_02	MLE-F-034	Multi-evaluation PDF covers all selected evaluations	Several evaluations saved.	Select several; download one report.	3 evaluations	Report includes every selected evaluation.	`test_charts_compare_reports::test_11_UT_02_multi_evaluation_report_covers_all_of_them`	As expected. 1 automated test(s) passed.	**Pass**
11_UT_03	MLE-F-034	No selection gives a message	Logged in.	Request a report with nothing selected.	Empty selection	Clear message; no file.	`test_charts_compare_reports::test_11_UT_03_no_selection_gives_a_message`	As expected. 1 automated test(s) passed.	**Pass**
11_IT_01	MLE-F-034	Report is built from saved data, not recomputed	Evaluation saved.	Generate the report twice.	Saved evaluation	Same content both times; model is not re-run.	`test_charts_compare_reports::test_11_IT_02_report_is_built_from_saved_data_not_recomputed` `test_charts_compare_reports::test_report_is_the_same_for_the_same_saved_evaluation`	As expected. 2 automated test(s) passed.	**Pass**
11_ST_01	MLE-F-034	Report refused when an evaluation no longer exists	Evaluation removed.	Request a report including it.	Deleted evaluation ID	Refused with a clear reason.	`test_charts_compare_reports::test_11_ST_02_report_is_refused_when_an_evaluation_no_longer_exists` `test_compatibility_at_upload::test_deleting_a_model_keeps_past_evaluations_readable`	As expected. 2 automated test(s) passed.	**Pass**

UC-12 Manage User Accounts

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
12_UT_01	MLE-F-040	Admin sees all accounts with status and role	Admin logged in; several accounts.	Open the user list.	3 accounts	Every account listed with role and status.	`test_admin::test_12_UT_01_admin_sees_all_accounts_with_status_and_role`	As expected. 1 automated test(s) passed.	**Pass**
12_UT_02	MLE-F-040	Admin can suspend and reactivate an account	Active Analyst account.	Suspend; then reactivate.	Analyst account	Status changes each time and the change is logged.	`test_admin::test_12_UT_02_admin_can_suspend_and_12_UT_03_reactivate`	As expected. 1 automated test(s) passed.	**Pass**
12_IT_01	MLE-F-043	Admin cannot suspend their own account	Admin logged in.	Try to suspend own account.	Own account	Refused; account stays active.	`test_admin::test_12_IT_01_admin_cannot_suspend_their_own_account`	As expected. 1 automated test(s) passed.	**Pass**
12_IT_02	MLE-F-043	Last active Admin cannot be suspended	Only one active Admin (another Admin is suspended).	Try to suspend the last active Admin.	Last Admin	Refused; an Admin always remains.	`test_admin::test_12_IT_02_last_active_admin_cannot_be_suspended` `test_admin::test_suspended_admin_cannot_use_admin_functions`	As expected. 2 automated test(s) passed.	**Pass**

12_ST_01	MLE-F-040	Invalid status request or unknown account is refused	Admin logged in.	Set a status the account already has; send an invalid status; target an unknown ID.	Same status / 'banned' / bad ID	Refused with a reason (404 for unknown account); nothing logged.	`test_admin::test_12_ST_02_setting_the_state_an_account_already_has_is_refused`
							`test_admin::test_status_must_be_active_or_suspended`
							`test_admin::test_unknown_account_returns_404`
								As expected. 3 automated test(s) passed.	**Pass**

UC-13 Manage User Roles

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
13_UT_01	MLE-F-041	Admin can promote an Analyst to Admin and back	Admin logged in; Analyst exists.	Change role to Admin; change back.	Analyst account	Role changes at once each time.	`test_admin::test_13_UT_03_admin_can_make_an_analyst_an_admin_and_back`	As expected. 1 automated test(s) passed.	**Pass**
13_IT_01	MLE-F-043	Admin cannot change their own role	Admin logged in.	Try to demote own account.	Own account	Refused.	`test_admin::test_13_IT_03_admin_cannot_change_their_own_role`	As expected. 1 automated test(s) passed.	**Pass**
13_IT_02	MLE-F-043	Last active Admin cannot be demoted	Only one active Admin.	Try to demote the last Admin.	Last Admin	Refused.	`test_admin::test_13_ST_02_last_active_admin_cannot_be_demoted`	As expected. 1 automated test(s) passed.	**Pass**
13_IT_03	MLE-F-042	Every change is logged with who, what and when; refused changes are not	Admin logged in.	Change roles and statuses; make a refused change; open the audit log.	Several changes	Log shows actor, target, old/new value and time; refused changes are absent.	`test_admin::test_13_IT_02_every_change_is_logged_with_who_what_and_when` `test_admin::test_refused_changes_are_not_logged` `test_admin::test_audit_log_page_lists_entries`	As expected. 3 automated test(s) passed.	**Pass**
13_ST_01	MLE-F-003, MLE-F-041	Only an Admin can create an Admin; first Admin comes from the command line	Empty database or Admin logged in.	Run create-admin; create an account as Admin and as Analyst.	create-admin command	First Admin is created; Analysts cannot create accounts; creation is logged.	`test_admin::test_first_admin_can_be_created_from_the_command_line` `test_admin::test_only_an_admin_can_create_an_admin_account` `test_admin::test_account_creation_is_logged`	As expected. 3 automated test(s) passed.	**Pass**

5.3 Non-Functional Test Cases

Performance (MLE-NF-001), reliability (MLE-NF-003), usability (MLE-NF-004) and extensibility (MLE-NF-005).

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
NF_PF_01	MLE-NF-001	Classification on 50,000 rows finishes within 10 s	Single user; model and 50,000-row dataset validated.	Evaluate and time the request.	50,000 rows, typical models	Completes in under 10 s.	`test_performance::test_classification_at_50000_rows_finishes_within_10_seconds`	As expected. 2 automated test(s) passed. Each run, including data set-up, took under 5.6 s (limit 10 s).	**Pass**
NF_PF_02	MLE-NF-001	Regression on 50,000 rows finishes within 10 s	Single user; model and 50,000-row dataset validated.	Evaluate and time the request; open the charts.	50,000 rows	Under 10 s; charts are sampled.	`test_performance::test_regression_at_50000_rows_finishes_within_10_seconds_and_charts_are_sampled`	As expected. 1 automated test(s) passed. Each run, including data set-up, took under 5.6 s (limit 10 s).	**Pass**
NF_PF_03	MLE-NF-001	Evaluation with a model close to 100 MB finishes within 10 s (manual)	A real ~100 MB model and a 50,000-row dataset.	Upload, validate, evaluate; time the request.	~100 MB model	Under 10 s, or the result is recorded as a known limit.	Manual (not automated)	As expected. Run on 4 Nov 2026: model 95.8 MB (1,112 trees), 50,000 rows, 5 evaluations took 1.59, 1.60, 1.62, 1.64, 1.67 s (limit 10 s; 5 of 5 within). Output saved in perf-100mb-output.txt.	**Pass**

NF_ER_01	MLE-NF-003	Malformed or non-object JSON is rejected clearly	Logged in.	Send an evaluation request as non-JSON, broken JSON, and a JSON list.	Bad request bodies	Each rejected with a specific JSON error.	`test_api_workflow::test_evaluation_request_must_be_json`
							`test_api_workflow::test_malformed_json_body_is_rejected`
							`test_api_workflow::test_json_body_that_is_not_an_object_is_rejected`
								As expected. 3 automated test(s) passed.	**Pass**

NF_ER_02	MLE-NF-003	Missing, blank or unknown evaluation inputs are named in the error	Logged in.	Send an evaluation with a field missing, blank, not text, and with unknown IDs.	Bad evaluation fields	Error names the field; unknown IDs give a clean 400.	`test_api_workflow::test_missing_evaluation_fields_are_named_in_the_error`
							`test_api_workflow::test_blank_and_non_string_evaluation_fields_are_rejected`
							`test_api_workflow::test_evaluation_with_unknown_ids_is_a_clean_400`
								As expected. 3 automated test(s) passed.	**Pass**

NF_ER_03	MLE-NF-003	Unknown route, wrong method and server errors return safe JSON	App running.	Open an unknown URL; use the wrong HTTP method; force an unexpected error.	Unknown path / wrong method / forced error	404 and 405 as JSON; 500 is generic and leaks no internal details.	`test_api_workflow::test_unknown_route_returns_json_404`
							`test_api_workflow::test_wrong_http_method_returns_json_405`
							`test_api_workflow::test_unexpected_server_error_returns_generic_500_without_leaking_details`
								As expected. 3 automated test(s) passed.	**Pass**

NF_ER_04	MLE-NF-003, MLE-SR-008	Rejected uploads leave nothing stored	Logged in.	Upload invalid files; then list models and datasets.	Invalid files	Lists contain no entry for the rejected files.	`test_api_workflow::test_nothing_is_stored_for_rejected_uploads`	As expected. 1 automated test(s) passed.	**Pass**

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
NF_US_01	MLE-NF-004	A new user reaches evaluation results unassisted (manual usability check)	New account; sample files; person who has not seen the app.	Ask the person to register, upload, validate, evaluate and open results without help; note where they hesitate.	sample_data folder	Completes the flow unaided. Any step that needs help is recorded as a defect.	Manual (usability check)	As expected. Run on 4 Nov 2026 with one classmate who is not on the team and had not seen the app. Finished register, upload, validate, evaluate and results unaided in about 2-3 minutes with no hesitation. No defects.	**Pass**
NF_US_02	MLE-NF-004	All pages render and expected form fields exist	Logged in.	Open every page; check forms and static files load.	None	Pages render; forms have the fields the script expects.	`test_app::test_workspace_pages_render_for_a_signed_in_analyst` `test_app::test_page_forms_have_the_ids_the_script_expects` `test_app::test_static_assets_are_served` `test_app::test_login_and_register_pages_render_without_an_account`	As expected. 8 automated test(s) passed.	**Pass**
NF_EX_01	MLE-NF-005	A new chart can be added by changing only the chart module	Source code available.	Register a new chart drawer and title; run the suite.	Test chart	No change needed in upload, storage or login code.	`test_charts_compare_reports::test_adding_a_chart_only_needs_a_drawer_and_a_title`	As expected. 1 automated test(s) passed.	**Pass**

6. Changes from the Earlier Test Plan

Area	What the system now does, and where it is tested
Scope of the plan	The earlier plan had 34 test cases for UC-01 to UC-05, all 'Not Executed'. This plan has 86 cases for UC-01 to UC-13 plus security and non-functional requirements.
Traceability	Every case names its SRS requirement (new MLE-F / MLE-NF / MLE-SR IDs); section 2 gives the reverse view.
Registration and roles (UC-01, UC-13)	Registration creates Analysts only. The first Admin is created from the command line and other Admins only by an Admin (01_*, SEC_02, 13_ST_01).
Data ownership (MLE-NF-002)	Each Analyst sees only their own data; other users' IDs return 404 (SEC_08 to SEC_12).
Compatibility (UC-04)	Also reported when a file is validated, not only at evaluation (04_IT_03).
Results (UC-07 to UC-11)	Confusion matrix, ROC, residual and predicted-vs-actual charts; saved history; comparison without ranking; PDF reports.
Admin (UC-12, UC-13)	Account list, suspend and reactivate, role change, audit log, last-admin protection.
Execution	Automated tests are mapped to each case. Results are recorded in BL-13.

7. Test Execution Summary (BL-13)

Item	Result
Date and time of run	4 November 2026, 20:50 IST
Environment	Windows, Python 3.14.6, pytest 9.1.1, run from the backend folder on the main branch
Command	python -m pytest tests -v -p no:cacheprovider --junitxml=junit.xml
Automated suite	219 tests run, 219 passed, 0 failed, 0 errors, 0 skipped, total time 56.9 s
Test cases in this plan	86: 86 Pass, 0 Fail, 0 not yet executed
Supporting tests	The suite also contains tests that are not listed against a case (for example hashing, e-mail case handling, duplicate-row counting). They also passed.
Performance (MLE-NF-001)	All three 50,000-row runs (logistic regression, random forest, regression) passed the 10 s limit. The manual case with a 95.8 MB model and 50,000 rows (NF_PF_03) also passed: five runs took 1.59 to 1.67 s.
Usability (MLE-NF-004)	One classmate who had not seen the app finished the whole flow unaided in about 2-3 minutes with no hesitation (NF_US_01). One tester only.
Defects found	None in this run.
Evidence	pytest-verbose.txt, junit.xml and perf-100mb-output.txt (docs/test-results)